

REDEFINING ACCURACY OF CRASH DETECTION SYSTEMS IN COMMERCIAL VEHICLES

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

by

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Under the guidance of

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November 2024

DECLARATION

I hereby declare that the thesis entitled **“Redefining Accuracy of Crash Detection Systems in Commercial Vehicles”** submitted by me, for the completion of the course “BECE497J – Project 1” to the school of electronics engineering, vellore institute of technology, vellore is bonafide work carried out by me under the supervision of **Velmurugan T.**

I further declare that the work reported in this thesis has not been submitted previously to this institute or anywhere.

Place : Vellore

Date :

Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled submitted by “**Redefining Accuracy of Crash Detection Systems in Commercial Vehicles**” Sumanyu Simhachalam(21BEC0381), Wakankar Ronit Ashish(21BEC0989), Raghavendran Ganesan(21BEC0996),SENSE,VIT, for the completion of the course “BECE497J – Project 1”, is a bonafide work carried out by him / her under my supervision during the period, . . 2024 to . .2024, as per the VIT code of academic and research ethics.

I further declare that the work reported in this thesis has not been submitted previously to this institute or anywhere.

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Internal Examiner

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Executive Summary

This project, *Redefining Accuracy of Crash Detection Systems in Commercial Vehicles*, addresses the critical need for faster emergency response in vehicular accidents by designing an advanced crash detection system. With increasing traffic-related hazards, rapid and precise crash detection can significantly improve survival rates by expediting emergency response.

Our system, based on an Arduino Uno microcontroller, utilizes an accelerometer to detect collisions and activate a GSM module. When an impact is registered, the system sends an automated message with the vehicle's GPS coordinates to a designated phone number, notifying emergency services. Additional sensors, including an ultrasonic sensor and sound sensor, enhance the system's ability to detect collisions accurately, further minimizing false positives.

The project integrates key hardware components—Arduino Uno, accelerometer, GPS and GSM modules—with critical software libraries, including SoftwareSerial for serial communication and TinyGPS++ for GPS data handling. The Arduino IDE facilitates code management, while FreeForm Apple aids in circuit design.

This crash detection system provides an efficient, automated solution for real-time accident reporting, which has the potential to save lives by reducing emergency response times and ensuring timely medical intervention.

In addition to the core functionality of crash detection and emergency alerting, our system is designed to be scalable and adaptable for a variety of vehicle types, including commercial fleets and personal vehicles. The modular nature of the hardware and software allows for easy integration of additional sensors or components, enabling further customization based on specific user needs. The system is also energy-efficient, designed to operate with minimal power consumption, which is crucial for maintaining functionality in power-constrained environments.

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List of Abbreviations

GPS	Global Positioning System
GSM	Global System for Mobile Communications
IoT	Internet of Things
SDG	Sustainable Development Goals
LoRa	Long Range
CAN	Controller Area Network Bus
SMS	Short Message Service
Wi-Fi	Wireless Fidelity

1.INTRODUCTION

Emergency response times have a significant impact on survival rates in car accidents, which are a major cause of death. In order to solve this, our project uses an Arduino Uno to create a crash detection system that uses an accelerometer to identify impacts. The technology automatically transmits the vehicle's location (latitude and longitude) to a selected phone number and a nearby emergency dispatch server in the case of an accident by turning on a GSM module. By giving early, reliable accident information, our approach improves emergency response efficiency, allowing ambulances to be dispatched more quickly and perhaps saving lives.

1.1. LITERATURE REVIEW

The development of crash detection systems has been driven by the need to reduce the high fatality rates associated with vehicle accidents, where the timeliness of emergency response is crucial to improving survival outcomes. In recent years, research has shown that rapid response to crash incidents can significantly increase the likelihood of saving lives and mitigating serious injuries. Traditional emergency response systems often rely on bystander reports, which can lead to delays in alerting emergency services, particularly in rural or remote areas. Consequently, automated crash detection systems that can independently detect accidents and alert responders have garnered significant attention in the field of automotive safety technology.

Current crash detection systems primarily rely on a variety of sensors to monitor vehicle dynamics. These include accelerometers, gyroscopes, and ultrasonic sensors, which together capture rapid changes in movement, orientation, and proximity to other objects. Accelerometers, in particular, have proven to be effective in detecting sudden decelerations that typically indicate collisions. Studies indicate that combining accelerometers with gyroscopic sensors allows for a more comprehensive analysis of movement patterns, providing a reliable means of differentiating between crashes and non-crash events, such as sudden braking or sharp turns.

In addition to accelerometers, ultrasonic sensors are sometimes used to measure the proximity of other vehicles or objects to detect potential collision risks. The integration of multiple sensor types enhances the reliability of crash detection systems, minimizing false positives and improving the accuracy of incident detection.

Research has also explored various methods of alerting emergency services in the event of an accident. GSM (Global System for Mobile Communications) modules are commonly used to

send real-time alerts to predefined contacts or emergency services, providing crucial information such as the vehicle's location. The integration of GPS modules further enhances the functionality by pinpointing the exact latitude and longitude coordinates of the accident site. This combination of GSM and GPS technologies has been widely studied, as it enables the rapid transmission of data to ensure timely intervention by emergency responders.

Some systems also include features like live tracking and direct communication with nearby hospitals, leveraging existing cellular networks to establish a fast and reliable communication link. Research supports that such systems can significantly reduce response times, especially in densely populated or high-traffic areas where precise location data is critical.

Despite the advancements in crash detection technology, existing systems often face limitations in terms of cost, power consumption, and complexity. Many commercial crash detection systems are embedded into high-end vehicles, making them less accessible for a wider population. Furthermore, power-intensive components can limit system reliability, particularly in low-power environments.

Recent research has also investigated the potential of machine learning algorithms in crash detection, allowing systems to learn from various types of driving and crash data to increase detection accuracy. Studies have shown that algorithms like decision trees, neural networks, and support vector machines can be trained to recognize patterns in sensor data that indicate potential collisions. By analyzing a wide range of driving behaviors and crash scenarios, machine learning-based systems can differentiate between typical vehicle movements and high-impact crashes, potentially improving reliability in more complex driving environments. However, these systems often require significant computational power and data storage, making them more suitable for advanced vehicles with onboard processing capabilities. This project, therefore, focuses on simpler, low-cost solutions that still deliver reliable crash detection without the need for complex algorithms or high-power resources.

This project aims to address these challenges by utilizing an Arduino-based system, which provides a cost-effective, flexible, and easily programmable platform for crash detection. The use of the Arduino Uno microcontroller enables seamless integration with sensors (e.g., accelerometer, ultrasonic, sound) and communication modules (GSM, GPS), allowing for efficient crash detection and notification without the need for extensive hardware. Additionally, software libraries such as SoftwareSerial, AltSoftSerial, and TinyGPS++ optimize data transmission, ensuring reliable communication between modules.

1.2. RESEARCH GAP

Despite significant advancements in crash detection systems, several critical gaps remain in the existing technologies, particularly in accessibility, affordability, and reliability in emergency notification. Many current systems are embedded into high-end vehicles or rely on expensive hardware, limiting their adoption among a broader population. This cost barrier restricts the deployment of crash detection technologies in everyday or budget vehicles, which constitute a large proportion of road users. As a result, lower-cost solutions are needed to make crash detection widely accessible.

Moreover, traditional crash detection systems often depend heavily on power-intensive components and complex integration processes, which can hinder their reliability in certain environments. Power limitations, especially in remote or rural areas, can prevent these systems from functioning optimally, affecting their ability to send emergency alerts in real-time. Research has shown that delayed or failed notifications can significantly impact response times, which are critical for survival in serious accidents. Consequently, a system that balances low power consumption with robust communication capabilities would fill an important gap in current designs.

Another gap lies in the limitations of sensor accuracy and false positive detection. Many crash detection systems, even those equipped with advanced sensors, struggle to consistently distinguish between actual crashes and non-emergency events, such as hard braking or sudden turns. This lack of precision often leads to false alarms, which can overwhelm emergency response services and reduce their overall efficiency.

Lastly, the lack of flexibility in software handling multiple communication modules simultaneously has been a bottleneck in traditional systems. Many systems rely on single-channel serial communication, which can restrict the simultaneous use of multiple sensors and modules.

By addressing these research gaps, this project seeks to provide an affordable, power-efficient, and accurate crash detection solution using an Arduino-based design that can seamlessly integrate multiple sensors and communication modules. This approach aims to enhance both the accessibility and reliability of crash detection technology, making it more practical for real-world applications.

1.3. PROBLEM STATEMENT

One of the major issues in road safety today is the lack of efficient and reliable systems for accurately detecting vehicle accidents and promptly alerting emergency services with precise location details. Current systems often rely on manual reporting, which can lead to delays in emergency response times. Such delays, especially with the increasing number of traffic accidents, can significantly impact survival rates, as every minute counts in delivering timely medical assistance to accident victims.

As the number of vehicles on the road rises, so does the frequency of traffic-related hazards, placing an even greater demand on emergency services. However, in many regions, there is no reliable automated system in place to rapidly detect accidents and relay essential information, such as exact GPS coordinates, to the nearest emergency responders. This lack of automation can result in slower ambulance dispatch and extended delays in getting accident victims the medical attention they urgently need, ultimately risking lives.

To address this critical gap, a solution is needed that not only detects accidents accurately in real time but also transmits essential details, such as the accident location, to emergency services without delay. This would enable faster ambulance response times, potentially saving lives by ensuring that victims reach medical facilities as quickly as possible.

Our project seeks to develop a crash detection system using accessible and cost-effective technology that provides an efficient means of accident detection, precise location reporting, and rapid communication with emergency dispatch, contributing to a more responsive and effective road safety ecosystem.

In addition to its immediate life-saving benefits, an automated crash detection system has the potential to contribute valuable data for traffic safety analysis and policy development. By collecting anonymized information on accident locations, frequencies, and patterns, such a system can provide insights into high-risk areas, common causes of accidents, and peak times for incidents. This data can be instrumental for governments and urban planners in designing safer roads, implementing targeted safety measures, and enacting policies that reduce accident rates over time. Thus, the system not only benefits individuals involved in accidents but also supports a safer transportation infrastructure for society as a whole.

1.3.1. RELEVANCE OF PROBLEM STATEMENT WRT TO SDG

1. SDG 3: Good Health and Well-being

The crash detection system contributes to SDG 3 by potentially reducing fatalities and serious injuries in road accidents. By enabling immediate alert notifications and providing precise location details, it supports rapid response from emergency services, minimizing delays in medical assistance and thereby increasing the likelihood of survival and reducing long-term injuries. This approach to accident management promotes safer road environments, aligning with the goal of improving health and safety standards for all.

2. SDG 11: Sustainable Cities and Communities

Under SDG 11, the goal is to make cities safer and more resilient. This crash detection system improves urban safety and infrastructure, providing a technological solution that enhances road safety and strengthens the emergency response framework. By integrating advanced technology such as GPS and GSM, it contributes to creating smarter, more responsive urban systems that benefit both individual drivers and the broader community, fostering sustainable and safe urban living environments.

3. SDG 9: Industry, Innovation, and Infrastructure

The crash detection system aligns with SDG 9 by promoting innovation in infrastructure and advancing industry standards for road safety. Through the integration of technology such as accelerometers, GPS, and GSM modules, the system demonstrates how cutting-edge solutions can improve transportation infrastructure. By fostering a safer and more resilient infrastructure, it contributes to building sustainable transportation networks that enhance connectivity, reduce the economic costs associated with accidents, and support sustainable industrial growth. This aligns with the goal of building resilient infrastructure and promoting sustainable industrial innovation.

2. PROJECT OBJECTIVE

The objective of our project is to develop a **Crash Detection System** using an **Arduino Uno** that can detect vehicle collisions and send real-time alerts to emergency services. The system aims to accurately identify sudden changes in vehicle movement, such as rapid deceleration or abnormal rotation, by utilizing sensors like **accelerometers**. This will enable the detection of crashes in real-time, allowing emergency services to respond promptly.

In addition to accurate crash detection, the system will be designed to be **cost-effective**, leveraging affordable components that make it accessible for a wide range of vehicles. By using **Arduino Uno** and common sensors, the system offers a budget-friendly alternative to existing high-end crash detection systems, making it more feasible for everyday use.

Beyond the initial crash detection and emergency alert capabilities, the system's modular design allows for future scalability and integration with more advanced technologies. By incorporating GPS and GSM modules, the system could provide exact location data to emergency responders, enhancing the precision and speed of the response. This modularity opens up opportunities to develop an interconnected safety network, where data from multiple crash detection systems can be collected and analyzed for broader road safety insights.

Moreover, this project demonstrates the viability of leveraging open-source hardware, like the Arduino Uno, to build critical safety solutions. The use of widely available components not only makes the system affordable but also adaptable for other vehicles and applications. The project can serve as a foundation for further innovations in automotive safety, promoting a culture of technological inclusivity by providing a low-cost safety solution that could eventually be implemented at a large scale across various types of vehicles.

The goal is to create a **working prototype** that can be further expanded, modified, or integrated with additional technologies, such as GPS and communication modules, to enhance functionality. Ultimately, the project aims to provide a practical, low-cost solution to improve emergency response times, contributing to greater vehicle safety and the potential to save lives.

3. PROPOSED WORK

3.1. SYSTEM MODEL

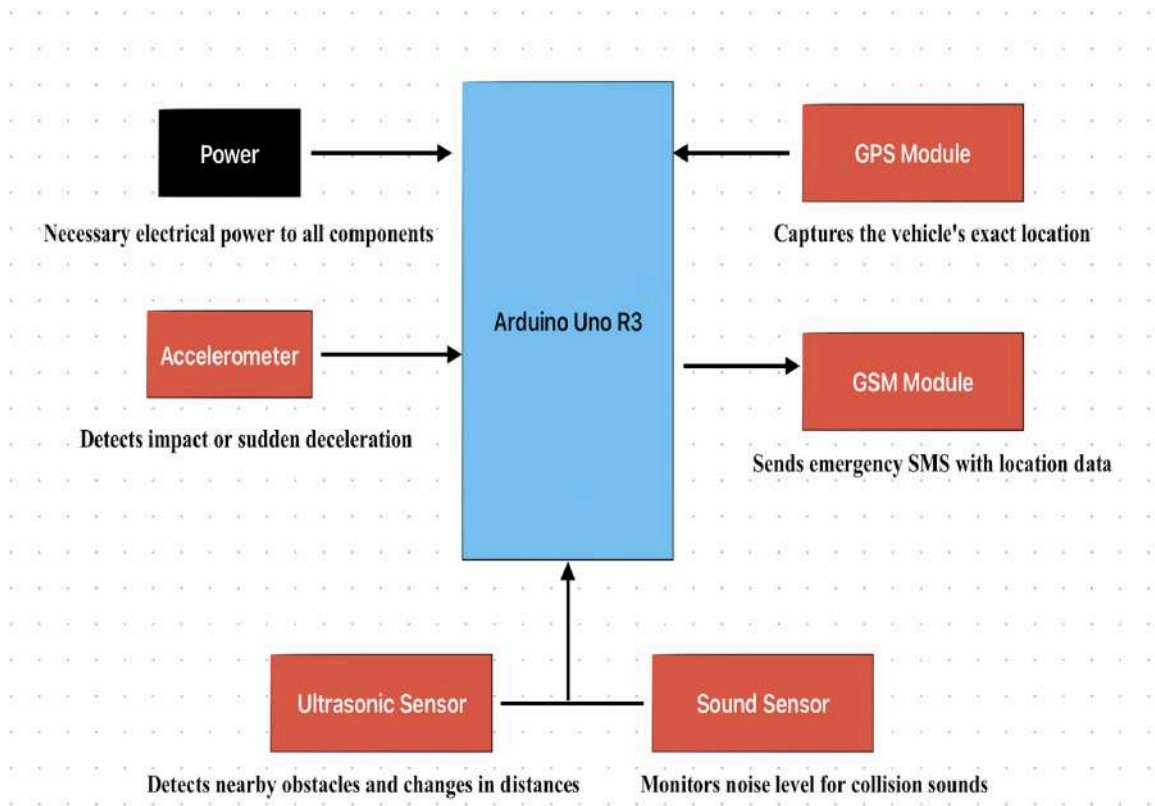


Figure 3.1.1 : System Model for Crash Detection System

The crash detection system is built around an Arduino Uno R3, which gathers and processes data from several key components. An **accelerometer** detects sudden impacts, while an **ultrasonic sensor** monitors the proximity of obstacles. Additionally, a **sound sensor** identifies loud noises that may indicate a collision. The **GPS module** captures the vehicle's exact location, and, in the event of an accident, the **GSM module** sends an emergency SMS with GPS coordinates to designated contacts, ensuring a quick response.

Together, these components allow the system to detect potential crashes and automatically notify emergency contacts, supporting timely emergency responses and improving road safety.

3.2. TECHNICAL DESCRIPTIONS

The System model illustrates a crash detection system using an Arduino Uno R3, integrating several key components to enhance vehicle safety. The system features a power supply that ensures all components receive necessary electrical power. The central unit, Arduino Uno R3, processes data from various sensors and modules. The ultrasonic sensor detects nearby obstacles by emitting ultrasonic waves and measuring the time it takes for them to bounce back, providing real-time proximity data to the Arduino.

The accelerometer plays a crucial role by detecting sudden changes in motion or impact, indicating potential crashes. Meanwhile, the sound sensor monitors noise levels, identifying sounds indicative of collisions. The GPS module captures precise location data, ensuring that the vehicle's location is accurately tracked.

In the event of an accident, the GSM module sends emergency SMS alerts, including the vehicle's GPS coordinates, to predefined contacts. This immediate communication ensures quick response times from emergency services. By integrating these components, the crash detection system is capable of real-time monitoring, analysis, and response. This comprehensive approach not only detects potential collisions but also provides crucial data and alerts, facilitating timely emergency responses and improving vehicle safety.

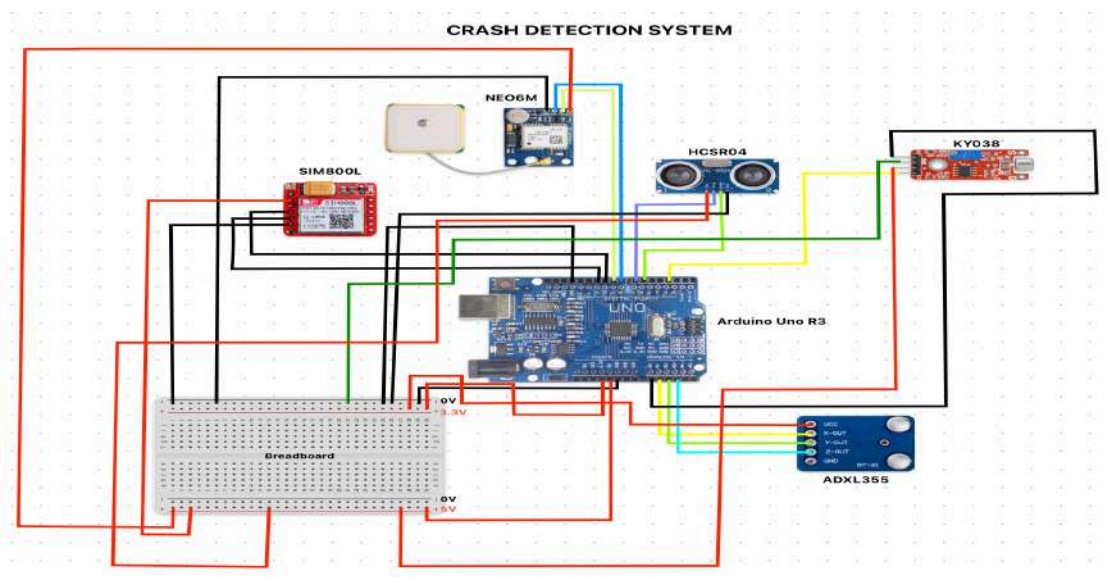


Figure 3.2.1 : Circuit Diagram of the Crash Detection System

4. HARDWARE/SOFTWARE TOOLS USED

Table 4.1: Hardware Components

Components	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller for processing sensor data and activating the GSM module during an accident.
Accelerometer	1	Detects impact force to identify potential collisions.
Ultrasonic Sensor	1	Provides real-time latitude and longitude coordinates of the vehicle.
Sound Sensor	1	Sends an SMS alert with the vehicle's location to emergency contacts in case of an accident.
GPS Module	1	Monitors nearby objects to enhance collision detection accuracy.
GSM Module	1	Detects sudden loud sounds associated with collisions, providing additional verification of a crash event.

Table 4.2 : Software Tools

Software Tool	Purpose	Details
Arduino IDE	Programming and Uploading Code	Provides a user-friendly interface for writing, compiling, and uploading code to Arduino boards. Converts code into machine-readable format and uploads it to the Arduino Uno via USB.
Freeform Apple	Circuit Diagram Design	Used for designing the system's circuit diagram. It is a flexible and user-friendly tool for visualizing connections between hardware components.

Table 4.3 : Software Libraries

Library	Purpose	Details
SoftwareSerial	Multiple Serial Communication	Allows the Arduino to create additional serial communication channels, enabling simultaneous communication with multiple devices like GSM and GPS modules using serial1.write() and gps.read().
AltSoftSerial	Enhanced Serial Communication	Provides better performance for serial communication at higher baud rates, minimizing errors in data transmission, essential for stable communication with GSM modules.
TinyGPS++	GP Data Parsing	Lightweight and efficient library for working with GPS data on Arduino. It simplifies reading and parsing GPS information, such as latitude and longitude.
Wire.h	I2C Communication with Sensors	Standard library for I2C communication, allowing the Arduino to interface with multiple devices using just two wires. Essential for connecting sensors like accelerometers, ultrasonic, and sound sensors.
Math.h	Mathematical Functions for Data Processing	Provides functions for handling complex calculations and signal processing, particularly useful for filtering and processing data from sensors. Includes trigonometric functions for analyzing trajectory and collision paths.

5. RESULT ANALYSIS

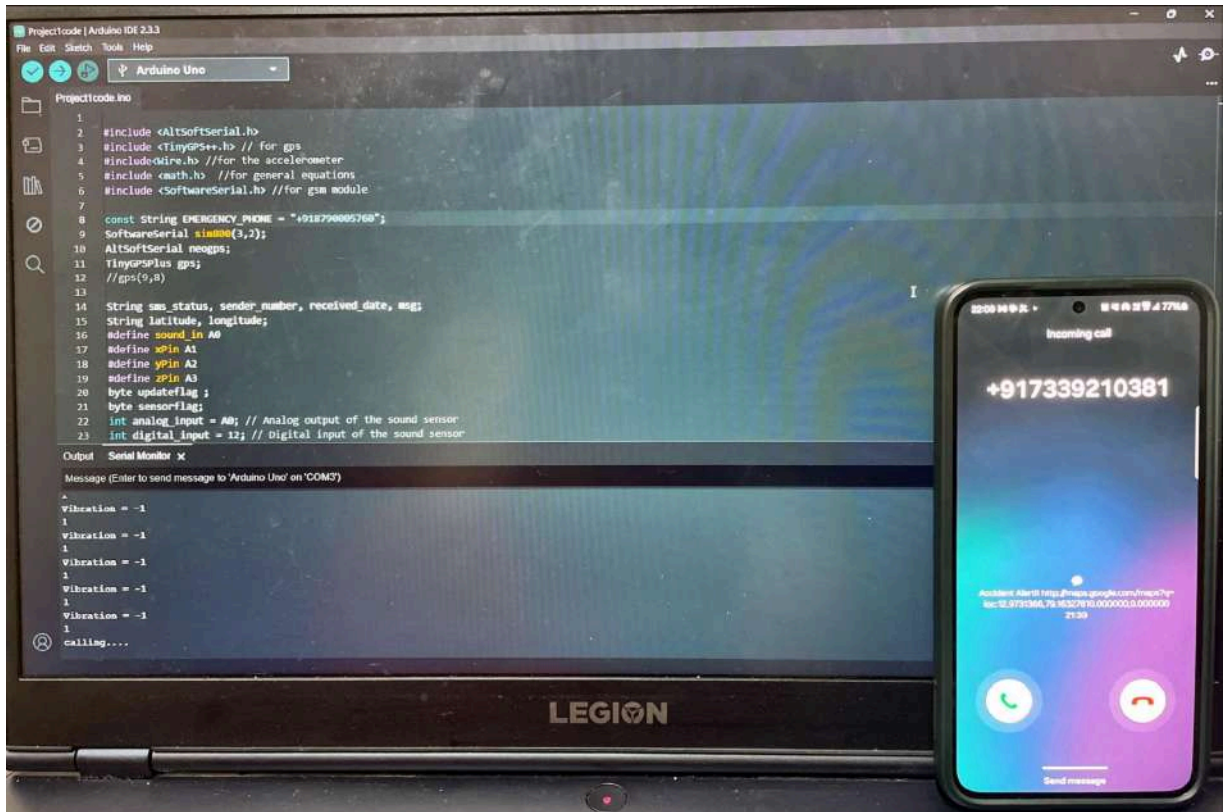


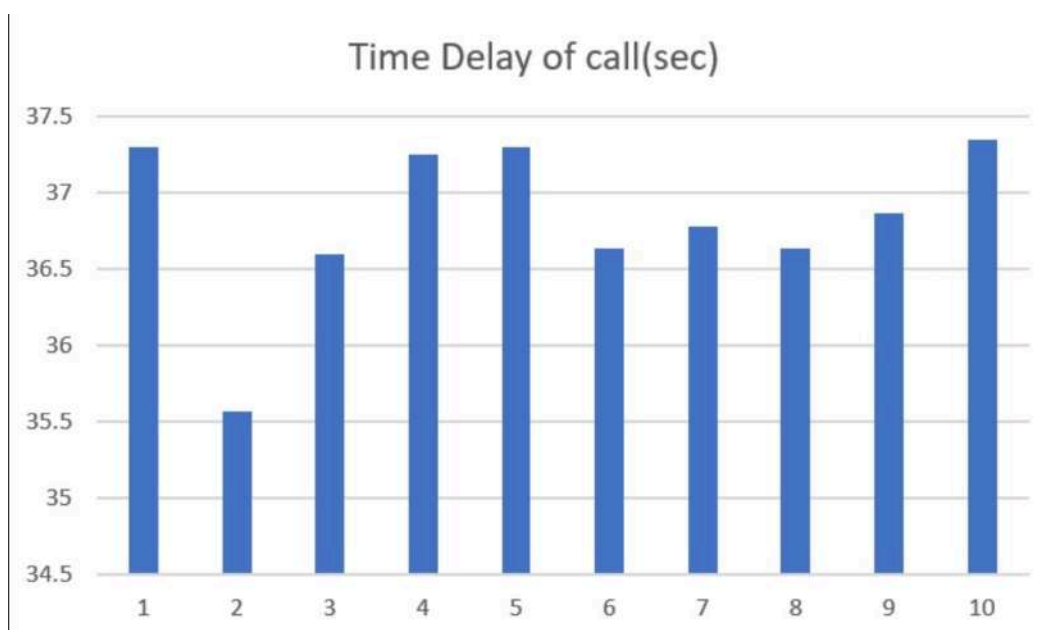
Figure 5.1 : Screenshot of the System's Response

From running tests on the crash detection system, we can observe some key points and drawbacks, such as:-

1. When two of the three sensors are triggered, the system is able to successfully interpret a crash.
2. The code runs smoothly.
3. Owing to the absence of a constant battery supply, the GPS module must be recalibrated every time we need to run a test. This recalibration takes a very long time.
4. The observed time delay between a crash detection and the call/message being sent is higher than we would have wanted.

RESPONSE DELAY ANALYSIS:**Table 5.1. Time Delay Between Impact Detection and GSM Alert across 10 Trials**

Trials	Time Delay(sec)
1	37.3
2	35.57
3	36.6
4	37.25
5	37.3
6	36.63
7	36.78
8	36.63
9	36.87
10	37.35

**Figure 5.2 : Delay Time Between Impact Detection and GSM Alert across 10 Trials**

Based on the data provided, here's an analysis of the time delay between the detection of a collision and the receipt of a call via the GSM module in the crash detection system:

1. **Consistent Delay Times:** The time delay across the 10 trials shows relative consistency, with values ranging from a minimum of 35.57 seconds to a maximum of 37.35 seconds. This consistency indicates a stable communication and processing time within the system.
2. **Mean Delay Calculation:** To provide a reliable average delay time, the mean delay across these trials can be calculated. Based on the provided data:
 - **Mean Time Delay** $\approx (37.3 + 35.57 + 36.6 + 37.25 + 37.3 + 36.63 + 36.78 + 36.63 + 36.87 + 37.35) / 10 = \mathbf{36.83 \text{ seconds (approx)}}$.
3. **Low Variation in Delays:** The variation between the minimum and maximum delay is approximately 1.78 seconds. This suggests that external factors, such as network conditions or processing speed variations, have minimal impact on the system's performance.
4. **Reliability of GSM Module:** The GSM module performs reliably, as there are no significant outliers or delays that exceed the normal range. This confirms that the system can be depended upon to trigger an emergency call within an average delay of approximately 36.83 seconds after a collision is detected.
5. **Potential for Further Optimization:** While the delay is reasonable, future improvements could aim at reducing this time further to enhance emergency response efficiency.

Overall, the results indicate that the crash detection system is efficient, with delays remaining well within a predictable range, which can be critical for timely emergency response in real-life crash scenarios.

6. CONCLUSION AND FUTURE WORK

6.1. SUMMARY

This project focuses on developing a crash detection system designed to improve emergency response times following vehicle accidents. The system utilizes an Arduino Uno R3 microcontroller integrated with a range of sensors, including an accelerometer, ultrasonic sensor, GPS module, and GSM module, to detect vehicle collisions and promptly notify emergency services. Upon detecting a crash, the accelerometer triggers the GSM module to send an alert message, including precise GPS coordinates, to a designated emergency contact. This immediate notification provides critical information to dispatch nearby emergency responders quickly, potentially improving the survival rate of accident victims. By leveraging accessible, low-cost hardware and efficient software, the system offers a practical and scalable solution that addresses limitations in existing crash detection technologies, such as high costs, power inefficiencies, and response delays. The project represents an effective approach to advancing vehicle safety and enhancing emergency responsiveness in accident scenarios.

6.2. LIMITATIONS AND CONSTRAINTS

The crash detection system has several limitations that need to be addressed for optimal performance. First, the sensors, particularly the accelerometer, may trigger false alarms due to sudden movements like hard braking or hitting a pothole, which can affect the system's reliability. Additionally, the system's dependency on continuous power could be a constraint, especially in environments with limited power sources or during prolonged use. The GSM module's effectiveness also relies on the availability of a stable mobile network, which may be problematic in remote or low-coverage areas. Similarly, GPS accuracy can be compromised in locations such as tunnels or dense urban environments, hindering the system's ability to provide precise location data in certain situations. While the system aims to be cost-effective, the use of low-cost components may result in reduced precision and durability compared to higher-end alternatives. The Arduino Uno R3's limited processing power also imposes constraints, particularly when handling multiple sensors and communication tasks simultaneously. Lastly, extreme environmental conditions, such as high heat, cold, or humidity, may degrade the performance and longevity of the sensors, further limiting the system's reliability.

6.3. IMPROVEMENT FUTURE WORK

While the current crash detection system demonstrates the potential to enhance emergency response times, there are several areas for improvement and future development. First, improving the accuracy of crash detection is essential. The system can be enhanced by integrating additional sensors, such as gyroscopes or pressure sensors, to help differentiate between actual accidents and non-crash events like sudden braking or sharp turns. This would reduce the occurrence of false positives and ensure more reliable notifications.

Another area for improvement is the power efficiency of the system. Integrating low-power components, optimizing the power consumption of the Arduino and sensors, and potentially incorporating a backup power system (such as a rechargeable battery) could ensure the system remains operational even in power-constrained environments. Additionally, using energy harvesting methods like solar panels could increase the system's sustainability.

The GSM module's dependency on network coverage can be mitigated by integrating alternative communication technologies, such as Wi-Fi, LoRa, or satellite-based communication systems, which would ensure the system remains functional in areas with poor cellular coverage. This would improve the reliability of emergency alerts, especially in rural or remote locations.

Future work can also focus on improving the GPS module's accuracy by using high-precision GPS systems, or incorporating a combination of GPS and local positioning systems (e.g., Bluetooth or Wi-Fi-based location services) to provide more accurate location data in areas with weak satellite visibility.

Furthermore, incorporating a more robust data processing system, such as a more powerful microcontroller or even a small embedded system, could improve the system's ability to handle multiple sensors and process data more efficiently. This would reduce delays and increase real-time response capabilities.

Finally, as vehicle connectivity advances, integrating the system with existing vehicular networks (e.g., CAN bus) could allow for more comprehensive data collection and integration, enhancing the crash detection system's overall functionality and providing real-time monitoring of the vehicle's health and status.

7. SOCIAL AND ENVIRONMENTAL IMPACT

The **Crash Detection System** proposed in this project has the potential to significantly impact both society and the environment. From a social perspective, the system improves road safety by providing timely alerts during accidents, which leads to faster emergency responses. This can save lives by ensuring that victims receive immediate medical attention. Additionally, the system promotes awareness of road safety and could encourage safer driving habits, benefiting both individuals and communities. By transmitting precise location data to emergency contacts, the system ensures that help reaches victims quickly, even in remote areas, making it accessible to a broader range of people.

On the environmental front, the system contributes to more efficient emergency response operations, potentially reducing the environmental impact of accidents. Faster response times could help alleviate traffic congestion, minimizing delays and lowering emissions from vehicles. The use of low-power sensors and modules also ensures energy efficiency, which is critical for vehicle-based systems. Furthermore, as the system evolves, there is potential for integration with other smart vehicle technologies, such as automated driving systems, which could contribute to more eco-friendly driving practices. Overall, the **Crash Detection System** not only enhances public safety but also supports sustainable transportation practices, making it a valuable innovation for both society and the environment.

Furthermore, the Crash Detection System has the potential to improve overall traffic management. By generating real-time data on accident locations and frequency, the system can provide valuable insights for city planners and transport authorities. This data can be used to identify high-risk areas, enabling targeted infrastructure improvements and more effective allocation of resources for road safety initiatives. Such insights can guide efforts toward building safer roads and implementing preventive measures, which could further reduce accident rates over time. In the long term, this system could help create safer, more resilient urban and rural road networks, benefiting society as a whole.

8. WORKPLAN

8.1. TIMELINE

The project, *Redefining Accuracy of Crash Detection Systems in Commercial Vehicles*, was carried out over a span of four months, from **July 2024 to November 2024**. In **July 2024**, the project began with the planning phase, where the scope and objectives were defined. During **August 2024**, design and development took place, including the selection and integration of hardware components, followed by the coding and initial testing of the system. In **September 2024**, further testing and refinement were performed to improve system functionality. **October 2024** focused on final integration and validation, where all components were assembled and the system was tested extensively. Finally, in **November 2024**, the project report was written, and the system was thoroughly tested and verified before the final submission.

8.2. INDIVIDUAL CONTRIBUTION

Sumanyu Simhachalam (21BEC0381) was primarily responsible for the **coding, compilation, and calibration** of key hardware components, including the **SIM800A GSM module, ADXL355 accelerometer, and ultrasonic sensor**. He also took charge of **regular maintenance** throughout the development process, ensuring that each component was functioning optimally and making adjustments as needed to improve performance and reliability. Sumanyu assisted with troubleshooting issues related to signal reliability and calibration accuracy, contributing to the system's consistent functionality.

Raghavendran Ganesan (21BEC0996) focused on **coding, GPS module calibration, and overall system testing**. He led efforts to ensure accurate location tracking and seamless data integration between the GPS module and other sensors. Additionally, he contributed significantly to the **documentation**, organizing and compiling the report, and preparing the project presentation..

Wakankar Ronit Ashish was responsible for **coding, designing the sequential logic, and calibrating and testing** the sound sensor to improve crash detection accuracy. He optimized the **sensor's response time** to detect loud sounds associated with collisions, contributing to the system's sensitivity. Ronit also provided valuable input in refining the sensor integration, ensuring all sensors functioned seamlessly within the system.

9. COST ANALYSIS

The cost structure of the Crash Detection System can be divided into initial setup, development, and operational costs. **Initial setup costs** include hardware components such as sensors (accelerometers, ultrasonic, and sound sensors), a GPS module, a GSM module, and the Arduino Uno R3, which are essential for crash detection, location tracking, and communication.

A key operational consideration for the crash detection system is the cost associated with purchasing and maintaining a SIM card for the GSM module. To enable real-time communication with emergency services, the system requires a SIM card capable of sending text messages or data over the cellular network. The initial cost includes buying a SIM card, which may be a one-time expense, depending on the region. However, ongoing costs involve periodic recharges or data plans to ensure sufficient balance for emergency alerts.

Development costs cover the design, coding, testing, and deployment phases, where integrating the hardware components and refining the system are critical. Once deployed, the system incurs **operational costs**, including **maintenance costs** for periodic sensor replacements, software updates, and bug fixes, which are relatively low compared to initial costs.

Additionally, **data transmission costs** arise from the use of the GSM module for sending alerts, with costs depending on the frequency of messages. As the system scales, particularly in larger fleets, both hardware and development costs increase, though economies of scale can help lower costs per unit. Despite the upfront investment, the long-term benefits of improved safety, faster emergency response, and potential insurance savings justify the costs of the system.

10. PROJECT OUTCOME

The **Crash Detection System** is designed to significantly enhance road safety by providing an automatic and immediate response in the event of a vehicle accident. Upon detection of a crash, the system uses a combination of **accelerometers, ultrasonic sensors, and sound sensors** to identify the occurrence and severity of the accident. This rapid detection allows for quick alerts to emergency services, significantly reducing the response time. The **GPS module** integrated into the system ensures that the exact location of the incident is provided, enabling first responders to reach the scene swiftly, even in remote or difficult-to-reach areas.

Additionally, the **GSM module** allows the system to send real-time SMS notifications to designated contacts, such as family members or emergency contacts, with precise location information. This ensures that the relevant parties are informed immediately and can take necessary actions, offering a higher level of safety for the vehicle occupants. By automating these processes, the system reduces the time between the accident and emergency response, potentially saving lives and minimizing the severity of injuries.

Faster emergency responses and the mitigation of accident severity can lead to fewer hospitalizations, lower insurance premiums, and reduced vehicle damage. Overall, the **Crash Detection System** promises to deliver significant improvements in road safety, reduce response times, and provide a cost-effective solution for both individual and commercial use.

❖ PUBLICATION/PATENT :

While this project has not yet resulted in a patent or publication, the innovative approach and design of the Crash Detection System present significant potential for future intellectual property development and scholarly publication. As the system evolves to incorporate additional functionalities, such as enhanced sensor integration, improved location accuracy, and energy-efficient components, pursuing a patent may become a viable next step to protect and commercialize the design. Additionally, a publication summarizing the system's design, implementation, and performance could contribute valuable insights to the fields of vehicular safety and IoT-based emergency response systems.

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